Lab 1

Due on 26 October

Select four datasets from the following repository:

`https://archive.ics.uci.edu/datasets`

- A regression dataset with a small training set (10^2<`training size`$<10^3$)
- A regression dataset with a larger training set (`training size`$>10^3$)
- A classification dataset with a small training set (10^2<`training size`$<10^3$)
- A classification dataset with a larger training set (`training size`$>10^3$)

For each dataset, set aside approximately 20% of the data points for validation.

1. For each of the two regression datasets, perform the following experiment.
   a) Sample training sets of increasing size (e.g. 10, 20, 30, etc.) For better results you can sample a training set multiple times for the same size. Do not use any data points in the validation set.
   b) Apply linear regression on each sampled training set and compute the mean squared error (MSE) of the learned model on the training set itself and on the validation set.
   c) Plot the MSE on the training set and on the validation set as a function of the number of samples.
   d) Explain the behaviour of the two curves.
2. Repeat the experiment with quadratic regression (i.e. linear regression with a non-linear quadratic transform).
3. Use cross-validation to identify which of the two models (linear or quadratic) is better for each regression dataset. How does the cross-validated performance compare to the performance of each model from 1. and 2. above?
4. Repeat the experiment for each classification dataset, but use logistic regression instead of linear regression and logistic loss instead of MSE.